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SESSION 2

Session 2 – Interactive Shell Main Loop and User Input Handling

Task 1: Create the main loop, display the prompt, read user input, handle exit conditions, design the control-flow diagram, and test the interactive

```
root@HP:~# nano session2.c
root@HP:~# gcc session2.c -o session2
root@HP:~# ./session2
myshell> hello
linux
operYou entered: hello
myshell> You entered: linux
myshell> ating system
exiYou entered: operating system
myshell> t
loop. root@HP:~#
```

```
root@HP:~# nano session1_exec.c
root@HP:~#
root@HP:~# gcc session1_exec.c -o session1_exec
root@HP:~# ./session1_exec
total 160
-rw-r--r-- 1 root root    0 Aug  2 05:29 demo1.c
-rwxr-xr-x 1 root root 16088 Aug  2 05:44 demo2
-rw-r--r-- 1 root root   516 Aug  2 05:44 demo2.c
-rwxr-xr-x 1 root root 16280 Aug  1 04:16 fork_example
-rw-r--r-- 1 root root   591 Aug  1 04:15 fork_example.c
-rw----- 1 root root    27 Aug  1 03:52 nano.867.save
drwxr-xr-x 2 root root  4096 Jul 31 18:23 neeha
drwxr-xr-x 2 root root  4096 Jul 31 18:23 neetu
-rwxr-xr-x 1 root root 16136 Aug  8 03:18 process
-rw-r--r-- 1 root root   706 Aug  8 03:16 process.c
-rwxr-xr-x 1 root root 16128 Aug 20 09:59 session1
-rw-r--r-- 1 root root   441 Aug 20 09:58 session1.c
-rwxr-xr-x 1 root root 16016 Aug 20 10:13 session1_exec
-rw-r--r-- 1 root root   215 Aug 20 10:13 session1_exec.c
-rwxr-xr-x 1 root root 16184 Aug 20 10:10 session2
-rw-r--r-- 1 root root   349 Aug 20 10:10 session2.c
drwxr-xr-x 3 root root  4096 Aug 20 09:57 shellforge
drwxr-xr-x 2 root root  4096 Jul 31 18:16 sri
-rwxr-xr-x 1 root root 16136 Aug  8 03:18 states
-rw-r--r-- 1 root root   576 Aug  8 03:17 states.c
-rw-r--r-- 1 root root    0 Jul 31 18:28 xls.txt
root@HP:~#
```

```
root@HP:~# ./session2
myshell>
You entered:
myshell> exit
root@HP:~#
```

Task 2: Capture keyboard input, handle Backspace, process the Enter key, manage the input buffer, support multi-character commands, and test user interaction.

```
root@HP:~# nano session2.c
root@HP:~# gcc session2.c -o session2
root@HP:~# ./session2
myshell> hellow
You entered: hellow
myshell> hello world
You entered: hello world
myshell>
You entered:
myshell> exit
root@HP:~# |

root@HP:~# nano session2_keyboard.c
root@HP:~# gcc session2_keyboard.c -o session2_keyboard
root@HP:~# gcc session2_keyboard.c -o session2_keyboard
root@HP:~# ./session2_keyboard
myshell> hello
hello
You entered: hello
root@HP:~# ./session2_keyboard
myshell> helllo
helllo
You entered: helllo
root@HP:~# |
```

Session 2 Observation:

The session focused on developing the interactive input functionality of the shell. The first task established the main shell loop, prompt, input reading, and exit mechanism, while the second task concentrated on keyboard-level input handling and buffer management. Through practical testing, the behavior of the shell during user interaction was observed and verified. The session provided a clear understanding of how an interactive shell receives, processes, and manages user input.